

Date	31 January 2025	Team ID	6ab61594fa5e5219dc387253
Project Name	Import Data using Transform Maps	Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II | ARCHITECTURE & STACK

Technology Stack

Technical Architecture: The data import and governance pipeline processes legacy CSV training records through staging tables, executing transform map coalesce logic, dot-walking manager references, and enforcing role-based ACL security.

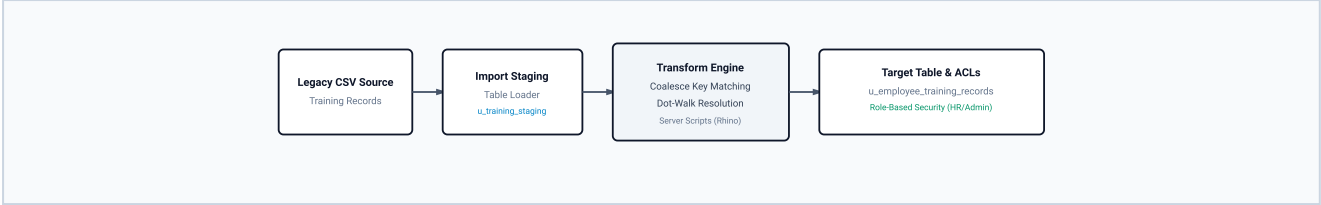


Table-1 : Components & Technologies

S.No	Component	Description	Technology
1	User Interface	ServiceNow workspace & admin forms for managing import sets and transform maps.	HTML, CSS, JavaScript, Jelly UI
2	Application Logic-1	Field mapping rules and transform map script execution for data transformation.	Server-side JavaScript (Rhino)
3	Application Logic-2	Reference resolution and parent manager lookup via dot-walking.	GlideRecord API / Dot-Walk Engine
4	Application Logic-3	Security enforcement ensuring confidential training scores are restricted.	Access Control Lists (ACLs) / Roles
5	Database	Relational schema for employee training records and lookup reference fields.	ServiceNow CMDB / Custom Tables
6	Cloud Database	Enterprise multi-tenant database backend hosting instance data.	ServiceNow Enterprise Cloud DB
7	File Storage	Storage handling for uploaded legacy training CSV data files and attachments.	ServiceNow Attachment DB / BLOB
8	External API-1	Inbound REST API endpoints for automated legacy data feed ingestion.	ServiceNow REST Table API
9	External API-2	Active Directory / LDAP integration for employee identity verification.	LDAP Integration Directory
10	Machine Learning Model	Automated duplicate record detection and data quality heuristics.	ServiceNow Predictive Intelligence
11	Infrastructure (Server / Cloud)	Enterprise PaaS deployment hosting the application instance securely.	ServiceNow Cloud Platform (PaaS)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source libraries utilized within the ServiceNow development framework.	AngularJS, Prototype.js, Rhino JS
2	Security Implementations	Role-based access control, table/field-level ACLs, and encrypted transport.	RBAC, ACLs, SHA-256, HTTPS/TLS
3	Scalable Architecture	Multi-tenant SaaS cloud architecture with asynchronous worker queues.	ServiceNow Job Scheduler / Node Clustering
4	Availability	High availability cloud infrastructure with automated failover and redundancy.	Distributed Cloud Server Clustering
5	Performance	Optimized transform batch processing and database indexing on coalesce fields.	Bulk Import Sets, DB Indexing, Caching

References

- ServiceNow Documentation: Import Sets and Transform Maps Guide (<https://docs.servicenow.com>)
- ServiceNow Developer Portal: Scripting and GlideRecord API Reference (<https://developer.servicenow.com>)
- ServiceNow Access Control Lists (ACL) Best Practices and Security Hardening
- IBM Cloud Architecture & C4 Model Standards (<https://c4model.com> / <https://www.ibm.com/cloud/architecture>)